Super Resolution On Satellite and Aerial Images using EDSR

***Abstract*—Super resolution is used to enhance spatial resolution in images using single low resolution image. In aerial and small satellite imaging quality of captured images is often constrained by camera's distance and surface of the Earth. The EDSR method is used in reconstructing high detailed imagery. High quality image is evaluated by PSNR and SSIM metrics. Experimental observation show that EDSR enhance image quality by retaining structural details. Also it reduces blurness compared to traditional interpolation methods. This approach can support better analysis of satellite images in the remote sensing applications. This paper explores on application of Single Image Super Resolution (SISR) methods, specifically EDSR, on the DOTA V1 which is a satellite and aerial imagery dataset, demonstrating improved image quality in terms of PSNR and SSIM.**

Keywords: SISR, Satellite and Aerial images, deep learning

I. Introduction

The images quality is often limited by cameras distance with surface of the Earth in aerial and small satellite images. The high cost and complexity of advanced cameras make them impractical for affordable small satellites with short lifespans. Atmospheric interference, sensor limitations, and optical distortions reduce image clarity. This makes it difficult to see small features like roads, vehicles or rooftops. These issues impact important tasks like object recognition, land use mapping and environmental monitoring, where fine details are crucial for precise analysis.

Methods like interpolation techniques (bilinear and bicubic) and image fusion have been used to enhance spatial clarity . However, these methods mostly increase image size without recovering fine details. It makes the image blur and loose fine details as well as edge sharpness .

Single Image Super Resolution (SISR) is an effective method to stated problems. It aims to convert a Fine resolution image from a single less resolution image [1] by restoring lost details and enhancing textures and edges.Unlike traditional methods (interpolation methods), SISR uses advanced computational models to fill in missing spatial information, greatly improving image quality.

Over time, several deep learning models have been created to improve image quality. It began with the CNN based approach and progressed to deeper, more refined networks. Methods like SRCNN, VDSR and EDSR gradually enhance the conversion of less resolution images to clearer, higher resolution versions.

SRCNN (Super Resolution Convolution neural networks): SRCNN is the first super-resolution technology using neural network model [2], It uses several convolution layers to convert less resolution images to higher resolution images . SRCNN had a straightforward design with just three layers. One layer extracts features from small patches, another transforms these features and the last layer reconstructs the high resolution image. This demonstrates that deep learning performs significantly better than older techniques like bicubic interpolation for adding details.

VDSR (Very deep Super Resolution): It extends the idea of SRCNN by a much deeper architecture. By increasing the network depth, VDSR can learn more complex mappings and recover finer details. VDSR also introduces residual learning, which helps mitigate the avoiding gradient problem and facilitates the training of deeper networks.[3]

EDSR (Enhanced Deep Super Resolution): EDSR further improves upon VDSR by removing unnecessary modules,which can consume significant memory and computational resources.[3] It is deep learning framework in image super resolution. It builds on the architecture of SRResNet. Its main innovation is removing batch normalization layers from the residual blocks. This change has been shown to improve performance and lower GPU memory use. The model simplifies the residual blocks and also increases the depth and width of the network by using more layers and channels.

MDSR (Multi scale deep Super Resolution) It is an improved version of EDSR that allows one unified network which handles sevaral upscaling factors .It uses a shared deep residual backbone along with specific pre processing and reconstruction modules by reducing computating cost and use of memory .

In the present work, our aim to use Enhanced Deep Super Resolution (EDSR) model to perform single image super resolution on aerial and satellite images and assess its effectiveness using Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index (SSIM) metrics. PSNR and SSIM are the most commonly used two metrics to understand the resemblance between original HR image and the super resolved image.

II. Literature Work

A. Trends in SISR

Fig 1 illustrates the yearly distribution of journal and conference publications indexed in IEEE related studied topic. An overall increasing trend is observed, indicating the growing interest in the image resolution field. The data reveals a clear publication trend that the conference papers were dominating between 2017 to 2019 and followed steady rise in journal publications after 2018. The observed trend reflects the evolution and increasing maturity of research in this domain.

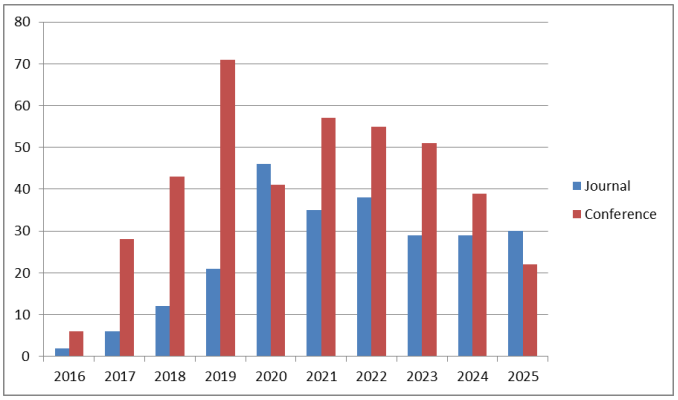


Fig. 1. Trends in SISR

B. Synthesis Table

TABLE I
COMPARISON OF SUPER RESOLUTION METHODS

Sl No	Method	Dataset	Reference	PSNR	SSIM
1	SRCNN	Set5	[1]	36.66	0.9542
		Set14	[1]	32.42	0.9063
		B100	[1]	31.36	0.8879
		Urban100	[1]	29.50	0.8946
		DIV2K	[1]	33.05	0.9581
2	VDSR	Set5	[4]	37.53	0.9587
		Set14	[4]	33.03	0.9124
		BSD100	[4]	31.90	0.8960
		Urban100	[4]	30.76	0.9140
3	EDSR	Set5	[5]	32.46	0.8976
		Set14	[5]	28.71	0.7857
		BSD100	[5]	27.72	0.7414
		Urban100	[5]	26.64	0.8029
		DIV2K	[1]	35.03	0.9695
		B100	[1]	32.93	0.9351
4	NAtSR	Set5	[5]	30.98	0.8606
		Set14	[5]	27.42	0.7329
		BSD100	[5]	26.44	0.6827
		Urban100	[5]	25.46	0.7602
5	MDSR	Set5	[5]	38.11	0.9602
		Set14	[5]	33.85	0.9198
		B100	[5]	32.29	0.9007
		Urban100	[5]	32.84	0.9347
		DIV2K	[1]	34.96	0.9692

In latest years, the single image super resolution field has noted a major shift from traditional interpolation techniques in deep learning based methods. As shown in table.1 , there has been consistent upward trend in performance matrices (PSNR/SSIM) correlating with increasing depth and complexity of neural networks.

Early deep learning models like SRCNN established the feasibility of learning end to end mappings from less resolution to higher resolution images. And then the trend moved towards deeper architectures like VDSR, EDSR and MDSR.

III. METHODOLOGY

The aim is to increase the spatial detail of the satellite and aerial images by using single image super resolution (SISR) . DOTA v1 dataset is used as it contains high resolution aerial images with small details like vehicles, roads, ships, buildings etc... Enhanced deep super resolution (EDSR) model is used and is trained on paired less resolution and higher resolution images. Low resolution(LR) images were generated from high resolution images by downscaling to 2x. The overall workflow can be split into 4 main parts.

1. Dataset preparation
2. Model architecture
3. Training process
4. Evaluation and performance assessment.

A. Dataset Preparation

To train model in supervised manner, paired lower resolution and higher resolution image set is required. A paired dataset of corresponding LR and HR images were created from the DOTA V1 high resolution aerial images. High resolution aerial and satellite images from DOTA were considered as ground truth. LR versions were created by down sampling each HR image by scale of 2 using bicubic interpolation.

For each training sample, 1024x1024 sized HR images and corresponding low resolution images with 512x512 size were used. Higher resolution image was degraded with gaussian blur of blur radius 2 and downsampled by scale of 2 to create low resolution image. The images were loaded as RGB tensors in the range [0, 255].

B. Model Architecture

EDSR is a deep learning model that converts a low resolution image to super resolved image by learning how to create the a sharper image during training. As shown in pipeline diagram, this implementation use scaling factor of 2.

This method starts with data preprocessing. To enhance training stability mean shift normalization is applied, this subtracts the mean RGB value from input image. This is to center the data distribution before entering deep learning layers.

After mean shift normalization, the image passes through a single convolution layer (3x3 kernel) in which it extracts features and highlight edges, textures and patterns. This layer expands the input from 3 channels to 64 feature maps which acts as initial set for residual body

The core component of the model is residual blocks . It consists of 16 identical residual blocks, which allows the model to learn complex details having high frequency. The image features are passed through two convolution layers and ReLU activation. the convolution layers mapping 64 input channels to 64 output channels. Every single residual blocks learns only the difference of LR image and original truth HR image. A scaling factor (0.1) is also applied to keep learning stable. The original input of the block is then reapplied to the processed output by using a global skip connection. This connection helps to learn the residual differences than full image mapping.

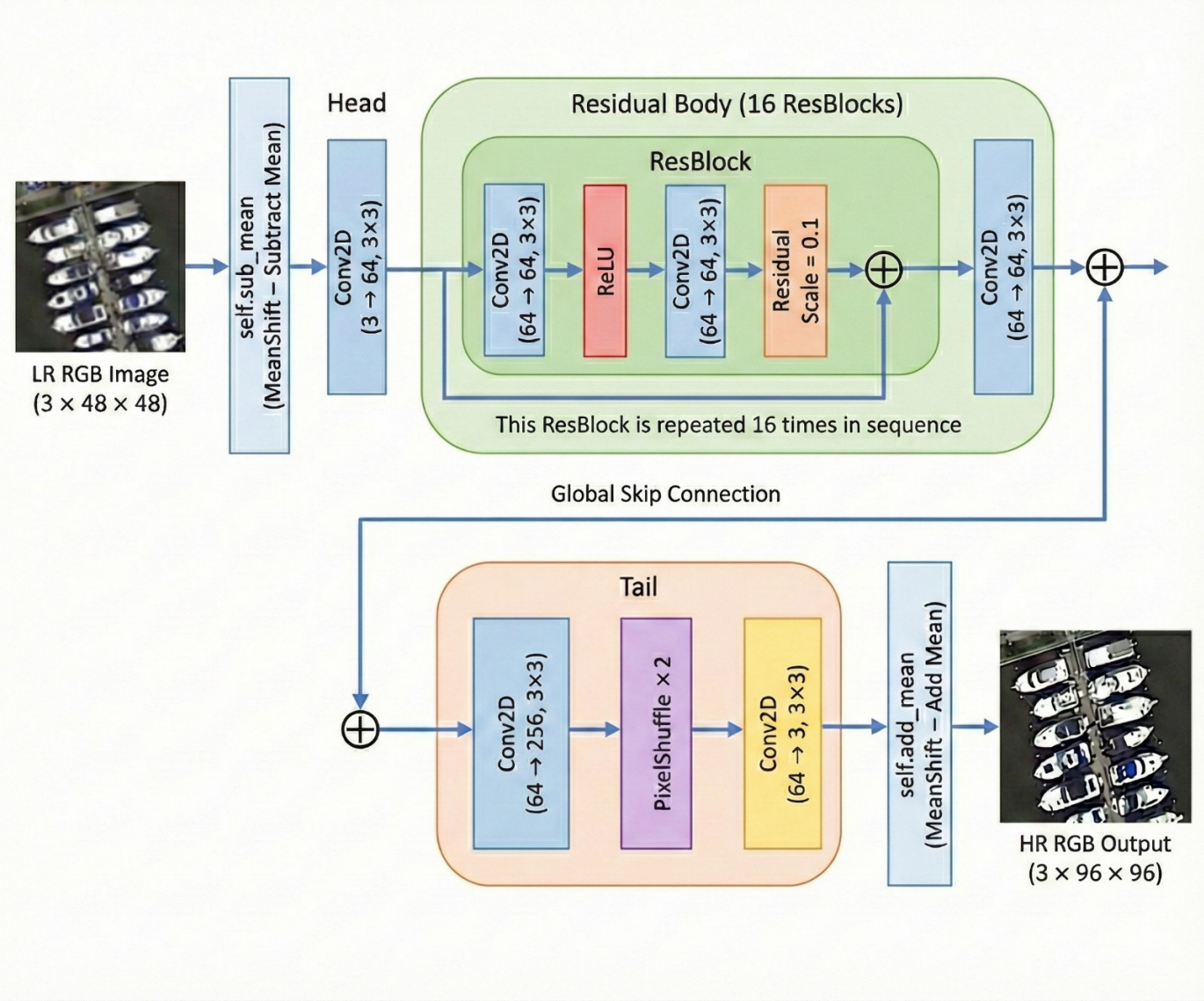


Fig. 2. EDSR Architecture

C. Training Process

Loss function: The EDSR model is trained using L1 loss function. L1 loss function in other terms called as Mean Absolute Error(MAE) and it is defined by

$$\mathcal{L}_{L1} = \frac{1}{N}\sum_{i=1}^{N}\left|I_{SR}^{(i)} - I_{HR}^{(i)}\right| \tag{1}$$

In this equation, I_{SR} and I_{HR} stands for super resolution image and ground truth image respectively. L1 loss is preferred over L2 loss because of better edge preservation, robustness to outlines, improved visual quality and stable and faster convergence [28].

Optimization: The optimization is done using Adam optimizer. The learning rate is taken as 1 x 10^{-4} and beta factors are set as $\beta_1 = 0.9$, $\beta_2 = 0.999$ .Gradient clipping applied with maximum norm of 1.0 to prevent exploding gradients. Automatic Mixed Precision (AMP) is used for faster convergence.

Training configuration of EDSR model:

TABLE II
TRAINING CONFIGURATION

Parameter	Value
Upscaling Factor	×2
Residual Blocks	16
Feature Channels	64
Batch Size	16
Epochs	100
LR Patch Size	48 × 48

D. Evaluation Metrics

In this method the image quality assessment(IQA) is done using PSNR and SSIM.

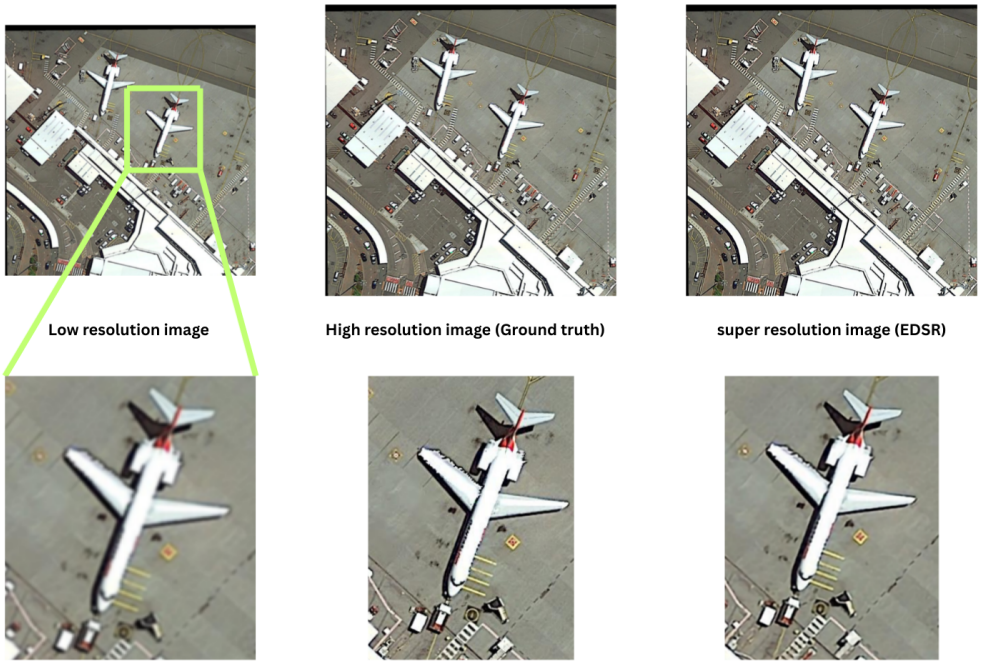


Fig. 3. Results: High resolution image obtained from EDSR method

1. Peak signal to noise ratio(PSNR):

It is the most commonly used image quality assessment method in the SISR field[7].It evaluates the quality of reconstructed image by comparing with the original HR image. PSNR is calculated using the mean square error (MSE) between two images and maximum possible pixel value[8].

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{\text{MSE}} \right) \tag{2}$$

If PSNR is higher then super resolved image will be close to the original high resolution image[7].

2. Structural Similarity Index Measure (SSIM):

It is the widely used image quality evaluation metric used to compare the similarity in two images with respect to their structure, brightness and contrast[8]. A higher SSIM value means the image is more similar to the original high resolution image and it preserves important structure details.

IV. Results and Analysis

The EDSR method was evaluated on DOTA v1 dataset which has high resolution satellite and aerial images. LR were generated from HR ground truth images using bicubic downsampling with scaling factor of x2. The metrics used for evaluation for SISR are PSNR and SSIM. PSNR measures the pixel level reconstruction accuracy, while SSIM performs on structural and perceptual similarity [6]. gig 4 shows the decrease in L1 loss with increased number of epochs. It indicates the effective learning and improved predictions over time. fig 5 shows the consistent upward growth of improvement in reconstruction quality with increase in number of epochs. It indicates that quality of reconstructed image improves as training continues. Overall the loss reduction and PSNR improvement demonstrate that the model successfully learns and gives higher quality outputs.

TABLE III provides the results of all the datasets trained on EDSR model. Our work on DOTA V1 yielded with 33.8499 PSNR and 0.8984 SSIM.

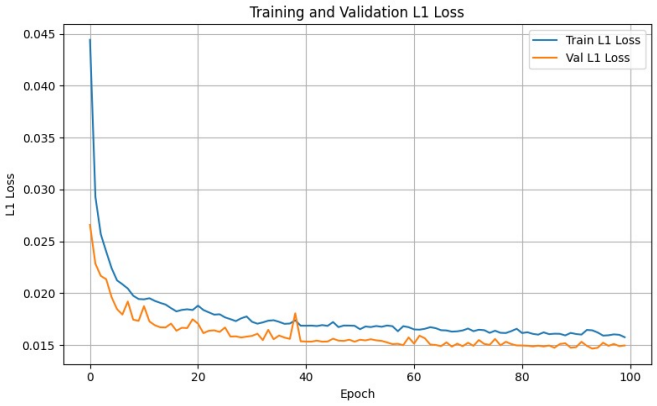


Fig. 4. loss graph

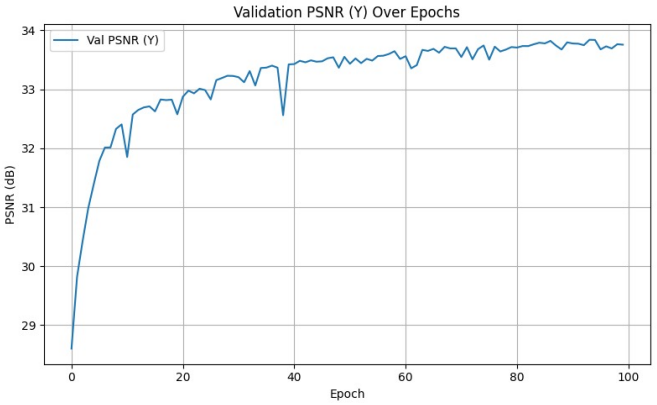


Fig. 5. graph showing PSNR validation over epochs

TABLE III
COMPARISON OF RESULTS WITH OTHER DATASETS IN SCALE OF 2X

	PSNR	SSIM
Set5[5]	31.775	0.929
Set14[5]	28.71	0.7857
BSD100[5]	27.72	0.7414
Urban100[5]	26.64	0.8029
B100 [1]	32.93	0.9351
DOTA V1	**33.8499**	**0.8984**

V. CONCLUSION

This paper focuses on application of Single image super resolution (SISR) techniques on the DOTA V1 dataset. The main aim was to increase the image quality and get more detailed finite images from their respective less resolution images. EDSR method proved to be a very good deep learning model as it focuses on learning important features in an image, leading to better reconstruction results. The DOTA v1 dataset was used, which contained high resolution aerial and satellite images making it suitable for the evaluation of super resolution performance. The results showed noticeable improvements in the detail and sharpness of the image, especially around edges and small objects. Compared to traditional upsampling techniques like bicubic, the EDSR model produce more accurate outputs. Although more super advanced super resolution models have been developed in recent years, EDSR remains strong and widely used as baseline.

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